BIP26 SENSATE-X 2.0

Student Getting Started Guide

Tenstar TS-ESP32-S3 | Waveshare LC29H(AA) GNSS | Arduino IDE

TU Dublin Tallaght Campus | June 2026

About This Guide

This is a short getting started guide covering the essential setup steps for the BIP26 lab hardware.

It is intended to help you get the equipment connected, the software installed, and your first program running.

A more detailed lab guide(s) with full exercises, explanations, and worked examples may be provided separately.

Work through the sections in order — each step builds on the previous one.

1. Installing the Arduino IDE

The Arduino IDE is the software you use to write, compile, and upload programs to the Tenstar ESP32-S3 board. It is free and runs on Windows, Mac, and Linux.

1.1 Download and install

1. Open a web browser and go to: https://www.arduino.cc/en/software
2. Under Arduino IDE 2, click the download link for your operating system (Windows, Mac, or Linux).
3. Run the downloaded installer and follow the on-screen instructions. Accept all default options.
4. When installation is complete, launch Arduino IDE from the Start menu or desktop shortcut.

[PHOTO / SCREENSHOT]

Screenshot: Arduino IDE download page at arduino.cc/en/software — highlight the Arduino IDE 2 download button

1.2 Add ESP32 board support

Arduino IDE does not include ESP32 support by default. You must add it through the Boards Manager.

5. In Arduino IDE go to File → Preferences.
6. In the Additional boards manager URLs field, paste the following URL:

Boards Manager URL

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

7. Click OK to close Preferences.
8. Go to Tools → Board → Boards Manager.
9. In the search box type ESP32.
10. Find the entry by Espressif Systems and click Install. This may take several minutes.
11. When complete, go to Tools → Board → ESP32 Arduino and select ESP32S3 Dev Module.

[PHOTO / SCREENSHOT]

Screenshot: Boards Manager showing esp32 by Espressif Systems with Install button

⚠ Important — Choose the Espressif Package

There are two ESP32 board packages available — one from Espressif Systems and one from Arduino.

Always install the one from Espressif Systems. This is the correct package for the Tenstar TS-ESP32-S3.

1.3 Set the correct board settings

After selecting the board, configure the Tools menu settings as shown below. These must be correct before uploading any sketch.

Tools Menu Setting	Value to Select
Board	ESP32S3 Dev Module
Flash Size	4MB (32Mb)
Upload Mode	UART0 / Hardware CDC
USB Mode	Hardware CDC and JTAG
USB CDC On Boot	Enabled
PSRAM	Disabled

[PHOTO / SCREENSHOT]

Screenshot: Arduino IDE Tools menu showing all settings correctly configured for ESP32S3 Dev Module

USB CDC On Boot: "Enabled"

CPU Frequency: "240MHz (WiFi)"

Core Debug Level: "None"

USB DFU On Boot: "Disabled"

Erase All Flash Before Sketch Upload: "Disabled"

Events Run On: "Core 1"

Flash Mode: "QIO 80MHz"

Flash Size: "4MB (32Mb)"

JTAG Adapter: "Disabled"

Arduino Runs On: "Core 1"

USB Firmware MSC On Boot: "Disabled"

Partition Scheme: "Default 4MB with spiffs (1.2MB APP/1.5MB SPIFFS)"

PSRAM: "Disabled"

Upload Mode: "UART0 / Hardware CDC"

Upload Speed: "921600"

USB Mode: "Hardware CDC and JTAG"

Zigbee Mode: "Disabled"

2. Installing the Required Libraries

Libraries provide ready-made code for communicating with the display, sensors, and GPS module. All libraries must be installed before compiling any BIP26 sketch.

2.1 How to install a library

12. In Arduino IDE go to Sketch → Include Library → Manage Libraries.
13. Type the library name in the search box.
14. Click Install next to the correct library.
15. If a dialog appears asking Install all or Install without dependencies — always choose Install All.

ℹ What are dependent libraries?

A dependent library is a helper library that another library uses internally.

For example, the BMP280 library needs Adafruit BusIO to handle I2C communication.

You do not call these dependent libraries directly in your sketch, but they must be installed or compilation will fail.

Clicking Install All handles this automatically.

2.2 Libraries to install

Search For	Type	Purpose
Adafruit ST7735 and ST7789 Library	Direct install	TFT display driver
Adafruit NeoPixel	Direct install	Onboard RGB LED
Adafruit BMP280 Library	Direct install	Pressure and temperature sensor
TinyGPSPlus	Direct install	NMEA sentence parser
SensorLib	Direct install	IMU gyroscope and accelerometer
Adafruit GFX Library	Installed automatically	Graphics drawing (dependency of ST7789)
Adafruit BusIO	Installed automatically	I2C/SPI bus support (dependency)
Adafruit Unified Sensor	Installed automatically	Sensor abstraction (dependency)

3. Connecting the External Antenna

The Waveshare LC29H(AA) board has a small IPX (also called U.FL) connector for an external GNSS antenna. Connecting an external antenna significantly improves signal reception, especially when working indoors.

3.1 About the IPX connector

- The IPX connector is a very small push-fit coaxial connector, approximately 2 mm in diameter.
- It is located on the top edge of the Waveshare LC29H(AA) board near the corner.
- The antenna cable has a matching IPX plug on one end and a larger SMA or magnetic-base connector on the other.

[PHOTO / SCREENSHOT]

Photo: Close-up of the IPX connector on the Waveshare LC29H(AA) board — arrow pointing to the connector location

3.2 How to connect the antenna

16. Hold the LC29H board steady on the anti-static mat.
17. Take the antenna cable and align the small IPX plug directly above the IPX socket on the board.
18. Press straight down gently but firmly — you will feel and hear a soft click when it is seated correctly.
19. Do not twist or angle the connector — press straight down only.
20. Place the antenna near a window or route the cable so the antenna has a clear view of the sky.

⚠ Handle the IPX Connector with Care

The IPX connector is very small and delicate. Connect and disconnect it carefully.
Always press straight down — never pull the cable sideways as this can damage the connector.
To disconnect, use a fingernail or flat tool to lever the plug straight up — never pull by the cable.
A poor antenna connection is a common cause of no GNSS signal or very low satellite counts.

✓ Antenna Placement Tips

For best results place the antenna on a windowsill or as close to a window as possible.

The antenna needs a clear view of the sky — walls, ceilings, and metal objects block GNSS signals.

It may take 1 to 2 minutes after powering on for the receiver to acquire satellites — this is normal.

The PPS (Pulse Per Second) LED on the LC29H board flashes once per second when satellites are locked.

[PHOTO / SCREENSHOT]

Photo: Antenna connected to the IPX connector with cable routed to a window — show the full setup

4. Wiring the LC29H to the Tenstar Board

Three Dupont jumper wires connect the Waveshare LC29H(AA) GPS HAT to the Tenstar ESP32-S3. The colour coding below matches the BIP26 lab kit wires.

4.1 Connection table

Wire	Waveshare Pin	Tenstar Front Label	GPIO	Signal Direction
Black	Pin 6 — GND	GND	GND	Common ground
Yellow	Pin 8 — Pi TXD0 (LC29H transmits)	A1	GPIO 17 (RX)	LC29H TX → Tenstar RX
Blue	Pin 10 — Pi RXD0 (LC29H receives)	A0	GPIO 18 (TX)	Tenstar TX → LC29H RX

⚠ Common Mistake — TX/RX Wiring

TX always connects to RX on the other device — never TX to TX.

The front label (A0, A1) and the GPIO number (18, 17) refer to the same physical pin — just different naming conventions.

If NMEA data is not appearing in the Serial Monitor, try swapping the yellow and blue wires.

4.2 Pin naming conventions explained

The same physical pin on the Tenstar board has three different names depending on the context. This is a common source of confusion.

Front of Board	Back of Board	GPIO Number	Used as in BIP26
A1	17	GPIO 17	UART RX — receives from LC29H
A0	18	GPIO 18	UART TX — sends to LC29H
GND	GND	GND	Common ground

ℹ Multi-Function Pins

A0 and A1 suggest analogue pins — but these pins are configured as digital UART pins in the BIP26 sketch.

ESP32-S3 pins are multi-functional. The same pin can be used as analogue input, digital I/O, or UART depending on how it is configured in software.

4.3 Full wiring schematic

The schematic below shows all connections between the Tenstar ESP32-S3 and the Waveshare LC29H(AA). Use this as a reference when wiring up your hardware.

[HAND-DRAWN SCHEMATIC]

Hand-drawn schematic showing: Tenstar TS-ESP32-S3 with pins GND, GPIO17(RX/A1), GPIO18(TX/A0) connected by three wires (Black/Yellow/Blue) to Waveshare LC29H(AA) pins 6(GND), 8(TXD0), 10(RXD0). Include USB-C power on Tenstar. Show IPX antenna on LC29H. Label all pins clearly with both GPIO numbers and front-label names.

[PHOTO / SCREENSHOT]

Photo: Completed wiring — both boards connected with three jumper wires, antenna attached, board powered on

5. Uploading Your First Sketch

Once the hardware is connected and libraries are installed, you are ready to upload a sketch to the board.

5.1 Open the sketch

21. In Arduino IDE go to File → Open.
22. Navigate to the BIP26_GNSS_DisplayV3.ino file provided by your lecturer.
23. The sketch will open in a new window.

5.2 Select the port

24. Connect the Tenstar board to your laptop using the USB-C cable.
25. Go to Tools → Port and select the port that appears (e.g. COM6 on Windows).
26. If no port appears, unplug and replug the USB cable and check again.

⚠ COM Port Changes are Normal

The COM port number can change — COM5, COM6, COM7 are all normal.

If the port disappears after uploading, check Tools → Port again. The board may have switched to a different port number after rebooting.

You will hear a Windows device connect/disconnect sound when the port changes — this is your cue to check the port.

5.3 Upload the sketch

27. Click the Upload button (right-pointing arrow) in the toolbar.
28. Wait for Done uploading to appear at the bottom of the IDE window.
29. Press the RST button on the Tenstar board to start the sketch running.

> **⚠ Always Press RST After Uploading**
> After uploading, the sketch does not always start automatically — pressing RST restarts the board cleanly.
> If the IDE cannot connect to upload, try the boot mode sequence: hold BOOT, press and release RST, release BOOT, then click Upload.

5.4 What you should see

- The TFT display lights up showing the GNSS Constellation Monitor screen.
- A navy header at the top with the title and TU Dublin / SENSATE-X 2.0.
- Date and time in cyan — this shows Waiting for time fix until satellites are acquired.
- Latitude, longitude, and altitude — shows Acquiring GPS fix until a position fix is obtained.
- Colour-coded bars for each GNSS constellation showing satellites in view.
- The onboard RGB LED pulses red while waiting for a fix, then cycles through rainbow colours once fixed.

[PHOTO / SCREENSHOT]

Photo: TFT display showing the GNSS Constellation Monitor — before fix (Waiting for time fix message visible)

[PHOTO / SCREENSHOT]

Photo: TFT display showing the GNSS Constellation Monitor — after fix (satellite bars filled, location showing)

6. Anti-Static Precautions

The Tenstar board and in particular the TFT display are sensitive to electrostatic discharge (ESD). ESD damage is permanent — always observe the following precautions.

- Always work on the anti-static mat — place the mat on the bench before handling any board.
- Touch the anti-static mat before picking up a board to discharge static from your body.
- Hold boards by the edges — do not touch the TFT display surface or component pins.
- Do not slide boards across ordinary desk surfaces or clothing.
- Store boards in anti-static bags when not in use.

> **⚠ ESD Damage is Permanent**

A board with ESD-damaged TFT will appear completely dark — no backlight, no display — even though the GPS and other functions still work normally.

ESD damage cannot be repaired. The board must be replaced.

Static damage can occur without any visible spark — even small discharges from your fingertips are sufficient.

7. Quick Troubleshooting

Problem	Try This First
TFT display completely dark	Press RST. Check USB Mode and Upload Mode settings in Tools menu.
Serial Monitor shows nothing	Press RST. Confirm baud rate is 115200. Check USB CDC On Boot is Enabled.
No NMEA data from GPS	Check wiring: Yellow → A1 (GPIO 17), Blue → A0 (GPIO 18). Try swapping yellow and blue.
COM port not found	Go to Tools → Port and select the new port number. Unplug and replug if nothing appears.
Galileo shows zero satellites	Move antenna closer to window. Wait 2 minutes. Galileo count will appear as geometry improves.
Compilation error	Check all libraries are installed. Search for the missing library name in Sketch → Manage Libraries.
Upload fails — cannot connect	Use BOOT+RST sequence: hold BOOT, press RST, release BOOT, then click Upload.

A full lab guide with detailed exercises will be provided separately.